

SST SENTINEL

Engineering Specification

SS-001 | Revision 0.2 | October 2, 2026

Track 2 - PolarFire SoC Icicle Kit

Parallel protection decisions, fixed emergency authority, and a synchronized record of the evidence and response.

DOCUMENT STATUS	PROPOSED ENGINEERING BASELINE
Implementation	No hardware, synthesis, routed timing or campaigns claimed
Coverage	28 sections; hardware, RTL, firmware, software, records and tests
Ownership	PE: power / FPGA; ES: embedded / FPGA; shared review
Use	Internal implementation specification; separate contest report <=10 pages

Numerical targets and candidate components are deliberately distinguished from verified facts. Open decisions have named owners and closure evidence. Material AI assistance is disclosed in Section 28.

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Status: proposed engineering baseline. No hardware, synthesis, routed timing, or campaign results are claimed.

This document converts the adversarial review and subsequent diagram discussion into explicit implementation requirements. It is the internal design specification; the competition's final report has a separate maximum of ten pages. This specification supersedes conflicting architecture details in the earlier review draft. The earlier proposal and architecture PDF remain reference artifacts, not revised released designs.

Team roles: **PE** = user, power electronics and shared FPGA implementation; **ES** = teammate, embedded software/logic and shared FPGA implementation; **BOTH** = joint review and integration. Registered team name and member initials remain open.

Document conventions

Label	Meaning
SHALL	Proposed mandatory behavior for this revision; verification required before acceptance.
TARGET	An intended numerical performance level, not a demonstrated result.
CANDIDATE	A proposed component, routing, or implementation that may change after review.
EXAMPLE	A calculation illustrating sizing; not an authorized physical operating limit.
OPEN	A value or decision that cannot honestly be closed with available information.
FACT	A manufacturer/platform or supplied-rules fact supported by the listed sources.

Requirements are identified by subsystem: SYS, PWR, AFE, IO, CLK, PRO, DSP, REC, MSS, SW, REG, EVT, TST, DOC. Verification methods are A (analysis), I (inspection), S (simulation), H (hardware measurement), and C (campaign). An OPEN acceptance limit requires team approval and a recorded rationale before its associated hardware test; an OPEN field is not silently assumed to pass.

1. Purpose and intended contribution

SST Sentinel shall demonstrate protection decisions based on parallel threshold timing and real switching context, with fixed emergency limits and an auditable record of each response. The application motivation is SiC converters in emerging 800 VDC power architectures and possible medium-voltage solid-state transformer cells. The prototype is a 48 V laboratory surrogate.

The central experiment is whether added context improves legitimate-transient acceptance at matched fault-intervention performance compared with competent fixed-threshold protection. Current slope alone is not assumed to identify a fault. Identical measured histories cannot be separated by arithmetic; ambiguous observations shall cause conservative shutdown.

ID	Requirement	Owner / verification
SYS-001	Use Track 2 and implement the protection, recording and connected viewer on the provided PolarFire SoC Icicle Kit or organizer-authorized substitute.	BOTH / I,H

ID	Requirement	Owner / verification
SYS-002	Build one protected half-bridge with two physically distinct current observations and two recorded voltage observations.	PE / I,H
SYS-003	The prototype shall use nominal 48 V; actual voltage/current/pulse/temperature limits shall be frozen in the approved operating envelope.	PE / A,H
SYS-004	Retain independent device-level gate-driver protection and fixed emergency limits in every physical power test.	PE / I,H
SYS-005	FPGA deadlines and emergency outputs shall not depend on Linux, network traffic, DDR service or event storage.	BOTH / S,H
SYS-006	Ground-truth fault/load trigger labels shall be excluded from classifier features. PWM generated by the actual controller is permitted context.	BOTH / I,S
SYS-007	Report accepted benign events, fault interventions and unresolved decisions separately; safe ambiguity is not presented as correct diagnosis.	ES / I,C
SYS-008	No MV qualification, universal fault separation, whole-system SEU immunity or SiC survival guarantee shall be inferred from this prototype.	BOTH / I

Scope tiers

Tier	Included
Required prototype	Emergency protection; bounded contextual decision; calibrated threshold sensing; four-channel local capture; bare-metal configuration; Linux Ethernet event viewer; physical fault/load campaigns and tuned baselines.
Target integration	Export completed events from SRAM to a reserved LPDDR4 queue; persist to eMMC; AMP isolation and Linux stress demonstration.
Optional after validation	Soft turn-off selection; calibrated Tang Nano/fast-DAC replay; richer oscillation diagnosis.
Future application	Isolated local nodes per MV cell, inter-cell coordination, 800 V hardware, EV/solar/traction adaptation.
Excluded dependencies	The separate 100 W AFE project, building an SST, an 800 V converter, machine learning, CAN and distributed MV control.

DDR export is part of the target system. Local SRAM capture is the first implementation milestone and keeps protection independent of DDR availability. Optional features shall not delay completion of the required prototype.

2. Numerical specification summary

Item	Baseline / status	Acceptance boundary
FPGA platform	Icicle / MPFS250T-FCVG484E, subject to supplied kit	Exact delivered revision and part checked before constraints/PCB release.
Fabric protection clock	TARGET 200 MHz	Routed timing and clock constraints pass; 5 ns tick granularity.
Emergency latency	TARGET less than 200 ns	Analog comparator-input threshold crossing to isolated driver inhibit; specified input/supply/load conditions.
Protection decision	TARGET at most 1 us	Analog warning crossing to command under qualified input conditions; digital qualified-warning latency reported separately.
Current sensing	2 distinct paths: I_dc, I_load	Current-path observability documented for every switching state.
Current comparators	3 per path; 6 outputs	Two calibrated programmable levels plus one independent fixed emergency level.
Recorded channels	I_dc, I_load, V_bus, V_sw	Common sample index; chip/analog skew measured.
ADC resolution / cadence	TARGET 16 bits; 1 MSPS per channel	Four serial lanes; acquisition/transport proven without missing frames.
ADC serial clock	CANDIDATE 25 MHz	Actual mode, setup/hold and framing checked.
Raw ADC payload	CALCULATED 8 MB/s	Four channels times two bytes times 1 MSPS; metadata is additional.
Sample ring	PROPOSED 128 KiB	16,384 frames of eight bytes each.
Event window	PROPOSED 8 ms pre + 8 ms post	8,000 preframes + 8,000 postframes = 128,000 raw bytes.
DDR queue	PROPOSED 64 MiB	Addressing, aliases, MPU/PMP and ownership validated.

Item	Baseline / status	Acceptance boundary
Capture counters	PROPOSED 64-bit fabric ticks and frame sequence	Atomic reading and wrap behavior specified.
Full J26 allocation	CANDIDATE 26 wired signal positions	Includes latch-clear and heartbeat omitted in the earlier draft.
Reduced J26 allocation	CANDIDATE 24 positions	Off-kit equipment supplies fault/load control; those two output pins are free.
Fabric planning allowance	$\leq 25,000$ LEs; ≤ 32 math; ≤ 80 LSRAM blocks	Planning limits, not utilization results.
Held-out physical trials	$\geq 1,000$ load steps; $\geq 1,000$ fault pulses	Valid preregistered shots; failures and abstentions counted explicitly.
Error-count target	Zero observed nuisance interventions and missed deadline-qualified faults	A target only; confidence and operating coverage reported.

The 1 us target concerns a bounded protection decision. It does not require complete oscillation diagnosis, multiple fresh ADC derivative samples, or device current extinction within that time.

3. System architecture and authority

The physical plant feeds the analog front end. Fixed emergency comparator outputs reach an external sticky trip and the FPGA. Programmable warning outputs reach FPGA threshold capture. The ADC feeds diagnostic math and event recording. FPGA logic controls complementary PWM, a global permission-to-switch output, and an optional global soft-profile request. External priority logic combines these with rail health, watchdog health and device protection. Linux displays and stores the event record.

Priority	Authority	Required behavior
1	External fixed-limit trip, device fault, rail failure, external watchdog failure	Force gate inhibit independently of a running processor; emergency profile wins.
2	FPGA sticky emergency or unsafe/expired decision	Revoke switching permission and latch the intervention.
3	Validated contextual decision inside bounds	May continue PWM or request a validated shutdown profile.
4	Bare-metal configuration and event service	Can disarm, configure and request arming; cannot override active faults.

Priority	Authority	Required behavior
5	Linux/dashboard/campaign host	Requests and observation only; no direct safety-register ownership.

All protection layers are part of the system. The external comparator path and FPGA path may share an amplifier or reference-related failure; this is a common cause, not redundant independent sensing. Retained gate-driver protection provides a different observable where supported. A formal functional-safety certification claim is outside scope.

4. Power stage, fault branch and operating envelope

4.1 Plant requirements

ID	Requirement	Owner / verification
PWR-001	Draw the actual converter topology and all fault-current, load-current and freewheel paths before component/PCB release.	PE / A,I
PWR-002	Use complementary gate commands with hardware deadtime and pulse-width/duty limits. Both switches shall remain off through reset/disarm/trip.	BOTH / S,H
PWR-003	A Microchip 700 V SiC MOSFET family is a candidate; select the exact package, gate voltage and driver together. Voltage rating is not the laboratory operating voltage.	PE / I
PWR-004	Place the controlled fault downstream of the protected power path so a verified action interrupts its source current.	PE / A,H
PWR-005	The crowbar branch shall have fault-rated R/L, independent maximum pulse duration and a defined commutation/energy-absorption path.	PE / A,H
PWR-006	The fault/load campaign host shall not extend the independent hardware pulse limit.	BOTH / S,H
PWR-007	Audit all source capacitance, load/filter capacitance, inductive energy and continued source delivery. Bench supply current limiting alone shall not be treated as an energy cap.	PE / A,H
PWR-008	Specify bleeder/discharge verification, charging interlock, shot cooldown and a repeat-shot inhibit when temperature or health exceeds approved limits.	PE / I,H
PWR-009	Prove that shutdown reduces the protected current below the approved safe-current limit for every physical test family.	PE / H,C
PWR-010	Record actual supply voltage, initial current, switching phase and temperature information for every campaign shot.	BOTH / H,C

The supply voltage range, maximum current, pulse width, repetition rate, temperature bins, inductance, resistance and device energy limits are OPEN. Selecting them requires actual components and lab equipment; the examples below shall not become limits by implication.

4.2 Calculated examples

Example	Calculation / meaning
DC-link capacitor	$0.5 \times 100 \text{ uF} \times 48 \text{ V squared} = 0.1152 \text{ J}$ in that capacitor only.
Initial inductive slope	$V/L: 48 \text{ V} / 1 \text{ uH} = 48 \text{ A/us} = 0.048 \text{ A/ns}$.
Higher-voltage slope	$800 \text{ V} / 100 \text{ nH} = 8 \text{ A/ns}$; this is an example, not a universal SiC fault law.
Bounded RL branch	48 V, 2.4 ohm, 4.8 uH: ideal final current 20 A, time constant 2 us, initial rise 10 A/us.
RL current at 3 us	$20 \times [1 - \exp(-3/2)] =$ approximately 15.5 A, for zero initial current and constant source voltage.
RL reach of 14 A	Approximately 2.41 us after onset; threshold latency and fault-onset latency are distinct.
Turn-off overshoot	$1 \text{ uH} \times 100 \text{ A} / 100 \text{ ns} = 1,000 \text{ V}$ before clamping and other circuit effects.

Capacitor droop, semiconductor drops, initial load current, parasitics and freewheeling modify the RL examples. An energy-limited output fault does not establish intrinsic SiC short-circuit survival.

4.3 Gate-driver contract

Driver selection is OPEN. The selected driver shall document logic voltage/polarity, input recognition thresholds, propagation delay, enable behavior, UVLO, local fault behavior, gate rails, Miller handling, retained device protection and the optional turn-off profile interface. Selection shall address the actual SiC package and source/Kelvin connections.

ID	Requirement	Owner / verification
PWR-011	External loss of permission shall inhibit both high-side and low-side gate outputs. Body-diode/freewheel current behavior shall still be measured.	PE / H
PWR-012	Emergency shutdown shall select the fastest physically validated profile, including overshoot constraints. 'Fast' is not an unconstrained maximum gate-sink current.	PE / H
PWR-013	Soft shutdown, if implemented, shall have a driver-side circuit or supported mode; establish setup/hold before revoke-permission.	PE / I,H
PWR-014	A single global soft request shall apply to both drivers; no independent per-device selection is claimed with this interface.	BOTH / I,S,H
PWR-015	Emergency logic shall force the validated emergency profile irrespective of the soft request. Verify concurrent soft/emergency transitions.	BOTH / S,H
PWR-016	Disable optional soft response until measured peak V_DS, gate behavior, current extinction and event energy pass approved limits.	PE / H

5. Analog sensing and comparator specification

5.1 Signal definitions and scaling

Signal	Physical role	Proposed representation / open limits
I_dc	Protected DC-path current; location finalized with actual loop diagram	Signed calibrated current; candidate shunt 5 milliohm and gain 10.
I_load	Distinct load-path current; must add information beyond I_dc	Same calibrated scale if shared DAC levels are used.
V_bus	Local DC-link voltage	Nominal 48 V; divider span, accuracy and protection OPEN.
V_sw	Switching-node voltage relative to defined local reference	Signed/offset span must include normal overshoot; divider/buffer ratings OPEN.
PWM state	Actual controller command and deadtime state	Fabric-generated state with sample-relative age; not a fault-injection label.
Independent health	Combined trip/driver/bus/watchdog/rail status	Fail-safe low means unhealthy; detailed external sources stored where observable.

R_shunt = 5 milliohm, gain = 10 gives 50 mV/A. At 20 A the shunt dissipates 2 W instantaneously and produces 100 mV before amplification; at 2 A it dissipates 20 mW. Pulse rating and layout remain mandatory checks.

For bipolar operation a candidate amplifier output offset is 1.65 V, giving 0.4 to 2.9 V over an illustrative -25 to +25 A span. This span/offset is a sizing example, not a frozen current limit. Confirm amplifier input common-mode, comparator range, supply headroom and ADC driver transfer. Unipolar operation may use a different offset only with a declared loss of negative-current observability.

ID	Requirement	Owner / verification
AFE-001	Each shunt shall use Kelvin sense routing; the schematic shall distinguish power return and sensor reference and prevent unintended current bypass.	PE / I,H
AFE-002	TARGET implemented current-sense bandwidth >=10 MHz at actual gain and load; measure differential steps, common-mode steps, settling and overload recovery.	PE / H
AFE-003	An amplifier shall not be placed across a switching common-mode outside its specified input range. Local isolation requires its own supply/return/delay design.	PE / I,H
AFE-004	Establish calibrated gain, offset, noise, drift, clipping limits and uncertainty over the actual test range. Acceptance error limit is OPEN until threshold margins are chosen.	PE / A,H

ID	Requirement	Owner / verification
AFE-005	Mark observations invalid after saturation, unsettled reference or common-mode disturbance until measured recovery criteria pass.	BOTH / H,S
AFE-006	Voltage dividers shall meet voltage/pulse/power requirements; clamping shall not disturb current-reference or DAC levels.	PE / I,H
AFE-007	Fast comparator branches and recording branches shall be separately checked so ADC input kickback/filtering does not alter emergency timing.	PE / H
AFE-008	The selected ADC's input common-mode, differential span and driver settling shall be implemented explicitly. A shunt-amplifier output is not automatically a compatible fully differential ADC input.	PE / I,H

OPA356 is a low-common-mode amplifier candidate only. Its offset can be consequential relative to a milliohm shunt: $9 \text{ mV} / 5 \text{ milliohm} = 1.8 \text{ A}$ input-referred error in an illustrative direct sense circuit before calibration. Selection requires an adequate measured residual error/drift budget. A gain-bandwidth number alone is insufficient. The 2.2 MHz AD8410A provides an example of why a conventional high-side amplifier may fail the fast-path bandwidth target. Sources: R08, R09.

5.2 Current threshold allocation

Channel	Lower warning	Upper warning	Emergency
I_dc	DAC A, through calibrated scale	DAC B, through calibrated scale	Fixed independent reference
I_load	DAC A, through matching calibrated scale	DAC B, through matching calibrated scale	Fixed independent reference

MCP4728 C/D are unassigned or available for reviewed self-test/calibration. No additional independently programmable levels are implied. Bus supervisor thresholds belong to the independent fault chain and consume no separate fast fabric comparator pins in this baseline.

ID	Requirement	Owner / verification
AFE-009	Six comparator signals shall be individually observed by the FPGA; each current emergency signal shall also reach the external latch.	BOTH / I,H
AFE-010	Warning references shall satisfy lower < upper < fixed emergency after all scale/offset/reference uncertainty. Verify ordering in voltage and current units.	PE / A,H
AFE-011	DAC warning changes shall occur only while disarmed; settling and threshold validation shall precede arming.	ES / S,H
AFE-012	DAC failure, I2C failure or EEPROM state shall not raise the fixed emergency limit.	BOTH / I,H
AFE-013	Fixed references shall have independent health checks; their supply dependence and common causes shall be documented.	PE / I,H
AFE-014	Hysteresis and bounded qualification shall have declared pulse rejection and detection effects. No blanking interval is treated as free noise immunity.	BOTH / A,H

ID	Requirement	Owner / verification
AFE-015	A negative/reverse-current fault is not considered covered by positive thresholds unless explicitly tested. Driver protection and a separate declared detection mechanism shall cover the selected topology, or the envelope shall exclude it.	PE / A,H

Illustrative thresholds of 6 A, 8 A and 14 A may be used in a calculation fixture only after independent hardware limits are reviewed. Actual thresholds are OPEN. Shared DAC levels are valid only when both current chains have matched calibrated mappings; otherwise add DAC channels and revise the design.

5.3 Comparator candidate and calibration

TLV3501/TLV3502 are candidates for CMOS-output threshold timing. The 12 ns allocation is conditional on the datasheet's stated low-overdrive input-step conditions, not an arbitrary ramp guarantee. MCP6561-family typical 50 ns-class behavior does not substantiate the original universal ladder budget, and its rise/fall skew is not channel-matching accuracy. Sources: R06, R07.

Threshold-time calibration shall vary input slew, overdrive, common-mode, supply, channel and temperature within available measurement capability. Store delay intervals rather than only a single correction constant. Record comparator direction and hysteresis. Verify threshold order against a scope or calibrated source independently of the FPGA.

6. Digital interfaces and pin allocation

6.1 Connector rules

The following is a candidate physical J26 assignment, not released package constraints. The reference design routes several MSS functions through fabric pins; those functions must be deliberately reclaimed. Physical positions 7 and 31 are unconnected in the referenced table. Proposed 3.3 V logic requires schematic and measured VDDI confirmation. Pin 1 or 17 being a 3.3 V supply does not prove every input is 5 V tolerant. Sources: R02, R03.

The previous 24-signal draft omitted external latch-clear and an independent watchdog heartbeat. This specification assigns physical pins 27 and 28 to those functions. Full on-kit test control now uses all 26 wired signal positions. Releasing the two test-only outputs yields a 24-signal production-style interface with two free positions.

J26 pin	Proposed signal	FPGA direction	Logic / reset state
3	DAC_SDA	Bidirectional open drain	Released; pull-up to verified I/O rail.
5	DAC_SCL	Open drain	Released; support bounded clock-stretch timeout.
11	CMP_DC_LO	Input	1 when calibrated lower warning is exceeded.
13	CMP_DC_HI	Input	1 when upper warning is exceeded.
15	CMP_DC_EMG	Input	1 when fixed emergency is exceeded.
16	CMP_LOAD_LO	Input	Lower warning.
18	CMP_LOAD_HI	Input	Upper warning.

J26 pin	Proposed signal	FPGA direction	Logic / reset state
22	CMP_LOAD_EMG	Input	Fixed emergency.
23	ADC_SCLK	Output	Idle low in selected CPOL=0 candidate mode.
24	ADC_CONVST	Output	Low while acquisition is stopped.
26	ADC_CS_N	Output	High/inactive.
19	ADC_SDI	Output	NOP/low when no configuration transaction.
21	ADC0_SDO_A	Input	Chip 0 result A.
12	ADC0_SDO_B	Input	Chip 0 result B.
29	ADC1_SDO_A	Input	Chip 1 result A.
32	ADC1_SDO_B	Input	Chip 1 result B.
33	PWM_H	Output	Low; external default off.
35	PWM_L	Output	Low; external default off.
36	SWITCH_PERMIT	Output	Low; external inhibit dominant.
37	SOFT_PROFILE	Output	Low; validated emergency/default profile.
38	HEALTH_OK	Input	Low/unhealthy on open/unpowered line where circuit permits.
40	EVENT_SYNC	Output	Low; scope trigger pulse on accepted event start.
8	TEST_FAULT_CMD	Output, optional	Low; locally hardware-gated and time-limited.
10	TEST_LOAD_CMD	Output, optional	Low; test-only.
27	EXT_CLEAR_REQ	Output	Low; active-high bounded pulse while disarmed only.
28	FABRIC_HEARTBEAT	Output	Low/toggling only after valid initialization.

ADC0 is proposed to digitize I_{dc}/I_{load}; ADC1 digitizes V_{bus}/V_{sw}. Confirm actual ADC SDO pad mapping in the selected dual-output protocol. All listed reset levels require external resistors/logic to enforce them before FPGA I/O becomes valid. Logical RTL reset values are not sufficient during unpowered or configuration states.

ID	Requirement	Owner / verification
IO-001	Build a revision-specific table from J26 pin to net to package pad to bank to implemented port and constraint. Every assigned signal shall have one checked route.	BOTH / I,H
IO-002	Remove conflicting QSPI/UART/ID-I2C/GPIO reference routes before binding custom RTL.	ES / I,S,H

ID	Requirement	Owner / verification
IO-003	Keep eMMC, Ethernet, console and required kit functions intact or document replacements. MikroBUS MSS SPI is not assumed to be a fabric fallback.	ES / I,H
IO-004	Specify I/O standard, drive, slew, pull resistors and timing constraints per signal group; verify cable/header ringing and loading.	BOTH / I,H
IO-005	No short header trace, input buffer or extra wire may be added without updated pin/timing budgets.	BOTH / I
IO-006	External trip clear shall be impossible while permission is asserted, and emergency set shall dominate clear. Select circuit/pulse timing before release.	PE / I,H
IO-007	Dedicated heartbeat shall drive an independent external watchdog; watchdog failure shall inhibit switching regardless of FPGA state.	BOTH / I,H
IO-008	Full allocation has zero spare signal positions. READY, ADC reset or extra driver-status pins require removing test outputs, changing the interface, or another verified route.	BOTH / I

ADC power-on reset and any recovery PD/RST handling shall be resolved at the AFE. A pulled-up reset pad is not assumed to provide software recovery. Off-kit/manual reset with switching inhibited, or a reviewed additional control path, is an OPEN decision.

6.2 External authority and heartbeat

Conceptual switching permission is the conjunction of valid FPGA permit, no sticky external trip, healthy driver/supervisor rails and healthy external watchdog. A fault input revokes both driver permissions. The selected isolator/driver shall default to inhibit if its primary input side is unpowered; verify polarity at both ends of the barrier.

CANDIDATE heartbeat period is 10 us and watchdog timeout is 100 us; these are health-management allocations, not the 200 ns emergency path. Freeze the actual timeout against the selected watchdog and operating envelope. Default startup shall remain inhibited until a valid heartbeat history exists. An independent fast emergency path remains necessary during the watchdog interval. The watchdog shall distinguish the expected disarmed startup state from an armed clock failure without granting switching permission accidentally.

7. Clock, reset and crossing specification

Domain	Candidate clock	Functions
Protection	200 MHz	Timestamps, warning rules, emergency latch, PWM/deadtime, feature validity, scope sync.
ADC serial	25 MHz derived activity	SCLK waveform and serial receive; prefer clock enables in a constrained fabric domain when feasible.
Diagnostic	100 or 200 MHz, OPEN	Sample calibration, window math and recorder frame handling.
MSS/FIC	Reference-design-compatible frequency, OPEN	Control/status transactions and DDR export.

ID	Requirement	Owner / verification
CLK-001	PF_CCC and reset sequencing shall hold switching inhibited until required clocks are locked and stable.	BOTH / S,H
CLK-002	Two-stage synchronizers shall handle asynchronous single-bit levels. Short emergency pulses shall be held by the external latch/health path rather than assumed to survive sampling.	BOTH / S,H
CLK-003	Multi-bit frames, configuration snapshots and event descriptors shall cross by coherent handshake or verified FIFO; independent per-bit synchronization is forbidden.	ES / I,S
CLK-004	Deassert resets synchronously in every clock domain; assertion shall revoke permission through the approved external/default path.	BOTH / S,H
CLK-005	CDC/reset reports and timing exceptions shall be reviewed. False paths shall be justified by actual synchronization structure.	ES / I,S
CLK-006	Calculate synchronizer MTBF with valid device data when available. Ordinary capture-phase latency is not a finite mathematical bound on metastability.	ES / A
CLK-007	Clock slowdown/stop shall not release or clear sticky trip. External watchdog/device protection shall remain independent.	BOTH / S,H
CLK-008	If 200 MHz timing does not close, revise tick units, deadlines, uncertainty and claims together before acceptance.	BOTH / A,S

The timestamp counter is a proposed free-running 64-bit tick counter. At 200 MHz one tick is 5 ns. ADC frame sequence is a separate 64-bit counter. Software shall use an atomic snapshot handshake; low/high register reads shall not produce torn timestamps.

8. Emergency path and latency contract

8.1 Measurement points

Point	Definition
AFE_TP1	Analog comparator input evaluated against its measured reference; capture the specified crossing/overdrive condition.
AFE_TP_C	Comparator digital output before FPGA/external branching.
AFE_TP_F	FPGA-generated revoke-permission output before external priority combining.
AFE_TP2_H / L	Isolated inhibit/enable recognition point at high/low driver.
AFE_TP_GH / GL	Device gate-source voltage using the appropriate local source reference.
AFE_TP_ID / VDS	Protected device current and drain-source voltage for physical intervention/stress.

Use labels distinct from the kit's own TP1-TP6. Desk/logic fixtures may evaluate the FPGA route while power switching is disabled; physical fault trials retain all backup protection. Simultaneously observe AFE_TP_F and local backup status so the backup cannot conceal a slow or missed FPGA action.

8.2 Emergency allocation

Stage	Allocation	Qualification
Comparator	12 ns	Candidate component bound under listed test conditions; ramps/near-threshold behavior measured.
Synchronization + registered trip	15 ns	Normal asynchronous phase at 200 MHz; metastability caveat.
Emergency digital qualification	Default 0 ns; permitted maximum 10 ns	At most two ticks; only if fault-energy and rejected-pulse evidence justify it.
Pad, PCB and external priority	15 ns	Allocation; actual part/load/layout OPEN.
Isolation	15 ns	Allocation; supply/load/driver-barrier details OPEN.
Total	67 ns with maximum qualification	Planning allocation, not measured worst-case performance.
Target	Less than 200 ns	Analog comparator crossing to driver inhibit within declared conditions.

ID	Requirement	Owner / verification
PRO-001	Any current emergency or independent health fault shall force revoke-permission and a sticky trip; neither software nor the classifier shall override it.	BOTH / S,H
PRO-002	Emergency input filtering shall default to zero additional digital qualification. A nonzero setting requires reviewed pulse/energy evidence.	BOTH / A,H
PRO-003	Measure both driver channels and each comparator path over phase, voltage and input edge conditions actually covered.	PE / H
PRO-004	Report fault-to-threshold, threshold-to-inhibit, inhibit-to-gate-off, fault-to-safe-current, peak V_DS and device event energy separately.	PE / H,C
PRO-005	Report median, p99, largest observed latency and measurement uncertainty; do not relabel largest observed as a universal worst-case guarantee.	BOTH / A,C
PRO-006	Any backup-assisted physical survival shall identify whether the FPGA itself met its action deadline.	BOTH / H,C

9. Warning capture, slope intervals and decision rules

9.1 Threshold timing

For each current path capture the first rising lower-warning tick and first subsequent upper-warning tick, with channel, direction, PWM state, edge-validity and calibration version. Missing, reversed, repeated/noisy and same-tick crossings receive explicit flags.

```
dt_observed = dt_true + differential comparator delay
               + route mismatch + capture uncertainty
dt_true is carried as [dt_min, dt_max]
dI is carried as [dI_min, dI_max]
slope interval = [dI_min / dt_max, dI_max / dt_min]
```

If $dt_min \leq 0$, the upper slope is unbounded/invalid for numerical division. A same-tick event may support a defensible lower bound using dt_max ; it shall not be converted into an arbitrary maximum slope. Threshold uncertainty is derived from shunt gain/offset, DAC/reference, comparator offset/hysteresis and calibrated input behavior. A 5 ns tick is granularity, not absolute analog accuracy.

ID	Requirement	Owner / verification
PRO-007	Calibrated interval comparisons shall require all contributing observations to be valid and current.	BOTH / A,S,H
PRO-008	Use multiplication/comparison or a checked bounded arithmetic method in the fast rule path; any divider latency shall appear in the deadline.	ES / A,S
PRO-009	Guard zero intervals, counter wrap, threshold chatter, absent edges and intervals below calibrated resolution.	ES / S
PRO-010	Cross-channel rules shall be derived from actual current-path equations for each PWM/freewheel state; no generic equality assumption is accepted.	PE / A,H

ID	Requirement	Owner / verification
PRO-011	Exclude observations contaminated by measured switching-edge disturbances or recovery intervals; exclusions cannot extend the decision deadline.	BOTH / A,S,H

9.2 Decision classes and action

Class	Meaning	Permitted action
EMERGENCY	Fixed limit or independent local/device/rail/watchdog fault	Immediate validated emergency shutdown; sticky.
BENIGN_COMPATIBLE	Available evidence satisfies the validated normal envelope with bounded uncertainty	Continue under unchanged emergency ceiling and bounded guard.
FAULT_COMPATIBLE	Valid observations support an abnormal path or unsafe progression	Shutdown; optional validated soft profile only with adequate physical margin.
UNRESOLVED	Insufficient, stale, contradictory or overlapping observations, or deadline expires	Conservative shutdown; sticky.

Oscillation and sensor-fault labels are later diagnostic outputs. A sensor failure requiring immediate protection shall enter HEALTH/UNRESOLVED/EMERGENCY as applicable; the longer diagnostic label does not postpone action.

Proposed rule precedence is: independent/emergency condition; invalid required evidence; current/time/energy guard violation; certified fault condition; certified benign condition; otherwise unresolved on expiry. Simultaneous conflicting certifications produce unresolved shutdown and a rule-conflict flag.

9.3 Observation window and freshness

ID	Requirement	Owner / verification
PRO-012	A qualified warning starts one hardware decision context with a fixed deadline. Retriggers cannot move that deadline later.	ES / S,H
PRO-013	TARGET decision command <=1 us after analog warning crossing under qualified input conditions. Report digital qualified-warning latency separately.	BOTH / S,H
PRO-014	CANDIDATE cycle allocation: capture allowance 8 ticks; observation <=150; rule/priority <=20; scheduling <=10. Total 188 ticks = 940 ns, leaving 60 ns allocation margin at 200 MHz.	ES / A,S,H
PRO-015	The capture allocation is conditional on comparator/qualification behavior. If analog capture exceeds it, close a revised budget or narrow the input envelope; do not silently reset the timing origin.	BOTH / A,H
PRO-016	Every feature shall carry age/validity. A sample-age limit and context mask shall be frozen per rule; OPEN limits shall disable that rule's benign authorization. The internal post-capture deadline shall be at most 180 ticks in this budget.	BOTH / A,S,H
PRO-017	The fast decision shall not await guaranteed fresh multi-sample ADC features. Available ADC history can inform context only with an approved age/error bound.	BOTH / A,S
PRO-018	A benign continuation shall have a finite guard/revalidation interval. Clearing the warning or reaching the deadline shall not erase an unsafe physical state.	BOTH / A,S,H

9.4 Physical guard

Before waiting or continuing, demonstrate at least:

$$I_{\text{warning}} + r_{\text{max}} \times (\text{remaining_decision_time} + \text{output_delay} + \text{physical_off_time}) + \text{current_uncertainty} < I_{\text{emergency}}$$

and separately validate allowable V_DS, fault energy and topology-dependent stress. An absent credible r_max does not permit waiting. A 1 us maximum window may need to terminate much earlier. I^2t is not a substitute for a SiC short-circuit thermal model.

The initial rule table, exact thresholds, slope intervals, current relationships, allowable feature ages, guard bounds and benign revalidation policy are OPEN design decisions. Freeze them only after calibration and development-set experiments, before held-out testing.

10. Protection state machine and rearming

State	Outputs / activity	Exit condition
RESET_INHIBIT	PWM off, permit low, soft low, clear low; external inhibit enforced	Rails/clocks valid; firmware initialized.
DISARMED	No switching; configuration writes permitted	Explicit self-test request or validated arming request.
SELF_TEST	Power path inhibited; calibration/stimulus checked	All required checks pass or FAULT_LATCHED.
READY	Inhibited; valid configuration/calibration and heartbeat health	Explicit arm accepted.
ARMED_NORMAL	PWM within duty/deadtime/current limits	Warning -> OBSERVE; any emergency -> FAULT_LATCHED; disarm -> DISARMED.
OBSERVE	Deadline runs; permission remains only while physical guard allows	Benign -> GUARDED_CONTINUE; unsafe/fault/expiry -> shutdown; emergency -> FAULT_LATCHED.
GUARDED_CONTINUE	PWM continues with finite validity/guard limits	Revalidation/return to normal, or shutdown on failure; deadline cannot be extended by chatter.
SHUTDOWN	Select permitted profile, revoke permission, capture reason	FAULT_LATCHED after action; emergency profile overrides at any point.
FAULT_LATCHED	PWM off, permit low, sticky reason and record retained	Explicit disarmed clear sequence with inactive sources, then repeat self-test/ready steps.

ID	Requirement	Owner / verification
PRO-019	Reset or illegal FSM state shall revoke permission and set an observable fault; use checked state encoding.	ES / S,H
PRO-020	Trip clear shall not arm, issue PWM, erase recorded event evidence or clear external faults that remain asserted.	BOTH / S,H
PRO-021	A clear command shall first ensure permit low, verify source health, issue bounded external-clear pulse with set dominance, verify clear, then require rearm authorization.	BOTH / S,H
PRO-022	No automatic restarting after a fault in the required prototype. Campaign rearm shall follow the same interlocked sequence.	BOTH / S,H,C

ID	Requirement	Owner / verification
PRO-023	Core heartbeat timeout, invalid required sensor evidence and configuration corruption shall produce a defined inhibit/fault response.	BOTH / S,H
PRO-024	Self-test shall cover DAC/reference ordering, ADC plausibility, comparator observation and output-path logic as practical. Lack of physical output coverage shall be stated, not inferred from register reads.	BOTH / I,H

11. ADC acquisition and diagnostic math

11.1 Converter transport

Two ADS9224R devices are CANDIDATE, each with two simultaneous channels. Their simultaneous conversion property does not establish zero inter-chip or front-end skew. Source R10 supports the device/interface choice; selected register codes and serial-mode timing remain implementation work.

ID	Requirement	Owner / verification
DSP-001	Acquisition shall target a common 1 us sample cadence for all four channels, with recorded aperture tick and frame sequence.	BOTH / S,H
DSP-002	Configure a dual-result-lane serial mode per ADC. Common configuration is permissible only when both chips require identical settings; independent settings need a revised CS/control plan.	ES / I,S,H
DSP-003	Candidate 25 MHz, 16-bit SDR transfer consumes 640 ns per lane before framing allowances. Verify the entire zone-2 sequence, not just bit-rate arithmetic.	ES / A,S,H
DSP-004	In zone-2 mode, associate returned data with the preceding sample's aperture timestamp. Pipeline fill after startup/recovery shall be flagged invalid.	ES / S,H
DSP-005	Input delay, board flight time, ADC output delay, FPGA setup/hold and selected launch/sample edges shall be constrained and measured.	BOTH / A,H
DSP-006	Malformed/stalled frames shall increment counters, mark sequence/validity gaps and invalidate dependent features. Do not insert silently fabricated samples.	ES / S,H
DSP-007	Long-window diagnostic failure or backpressure shall not stall threshold capture or deadline evaluation.	ES / S,H

11.2 Fixed-point and feature definition

Quantity	Proposed internal form	Explicit behavior
Raw ADC	Unsigned 16-bit code per channel	Exact transport bytes preserved.
Calibrated current	Signed 32-bit milliamps	Gain/offset conversion with documented rounding; saturation flagged.
Calibrated voltage	Signed 32-bit millivolts	Supports offset-mapped V_{sw} ; full scale/offset included in metadata.
Time	Unsigned 64-bit 5 ns ticks	Use modular-safe interval logic; software reads atomically.
I squared	Unsigned product of current magnitude	Use sufficient intermediate width; no wrap-to-small values.
I^2t window	Proposed N=1,024 samples; 80-bit sum in mA squared times sample count	Window 1.024 ms at nominal cadence; purpose/physical threshold reviewed; no SiC thermal inference.
ADC derivatives	Difference over stated sample spacing/window	Units and noise/age bounds explicit; not nanosecond slope.
Comparator slope	Calibrated current/time interval	Validity accompanies lower/upper bound; no zero-denominator value.
Cross-channel residual	Topology/PWM-specific equation	Sign conventions and delays calibrated.

Calibration multiplies can require several 18 x 18 math blocks for a 32-bit representation; the 32-block allowance is not an implicit one-block-per-operation promise. Freeze gain coefficient format, rounding and scaling in a golden reference model before RTL implementation. Any truncation that could turn an unsafe value into a safe one shall be prohibited or conservatively biased.

Proposed arithmetic ABI: gain coefficients are signed Q16.16; offsets are signed milliamps/millivolts; use widened signed 64-bit multiply/add before rounded 32-bit calibration output. Slope guards use unsigned Q16.16 milliamps per 5 ns tick. Implement interval products and shifted current differences with checked 64-bit intermediates; conversions to A/us are software presentation. Squaring a signed 32-bit current uses a widened nonnegative magnitude and a 64-bit product; the rolling 1,024-sample sum uses 80 bits. Store that sum and sample period separately; software converts to A squared seconds. For an exact mA-squared-ns integral, use a widened 96-bit product. Review all coefficients/ranges before ABI freeze.

Proposed recording-branch anti-alias target is approximately 250 kHz; exact filter/ADC driver response is OPEN. Keep the fast comparator branch separate. Candidate longer diagnostics use 256 samples (256 us) and a nominal 20-200 kHz oscillation-analysis range, subject to measured signal/filter/noise behavior. Such diagnostics are not a

universal oscillation detector. A derivative over a proposed 16-sample lag describes a 16 us change, not switching-edge dv/dt. Scope instrumentation remains the switching-peak measurement authority.

ID	Requirement	Owner / verification
DSP-008	Choose rolling/windowed or declared leaky I^2t behavior; indefinitely accumulating healthy current is prohibited.	BOTH / A,S
DSP-009	Overflow shall saturate conservatively and set a sticky diagnostic flag. Guard arithmetic shall not wrap.	ES / S
DSP-010	Define the minimum oscillation frequency, observation duration, sample count, hysteresis and alarm criterion before claiming oscillation detection.	BOTH / A,H
DSP-011	Establish sensor-stuck/open/improbable checks using commanded self-test and topology relationships; a quiet signal alone shall not prove a failed or healthy sensor.	BOTH / S,H
DSP-012	Store feature validity, age and the calibration/rule versions supporting each decision.	ES / S,H

12. Recorder, buffers, DDR and storage

12.1 Frame and timing model

Raw frame layout is eight bytes, little endian: bytes 0-1 I_dc, 2-3 I_load, 4-5 V_bus, 6-7 V_sw. All are raw ADC codes. Engineering scaling is supplied separately. The frame's sample sequence and aperture tick are inferred only over proven continuous cadence. Gaps shall be represented in a validity/gap record, not hidden by inference.

Allocate 128 KiB for 16,384 frames. Additional edge records, validity data and descriptors have separate budgets. A four-bit per-frame validity map costs 8 KiB if stored for the full ring; choose that or an equivalent reviewed sparse-gap encoding. Neither is included in the eight-byte raw payload.

At a trigger, identify the first aperture at/after trigger as postframe 0. Select the 8,000 preceding frames and 8,000 postframes. If initial prehistory is insufficient, record its actual length and a SHORT_PRE flag. After 8,000 postframes, freeze the sample ring for transfer. With 16,000 requested frames in a 16,384-frame ring, 384 frames remain as implementation margin; this is not a second capture bank.

12.2 Capture state machine

State	Behavior
WARMUP	Fill prehistory; preserve protection/diagnostic operation.
RING_READY	Continuously overwrite old frames; capture primary trigger metadata.
POST_CAPTURE	Continue exactly the selected postframe count; lock primary trigger origin and add bounded secondary edge metadata.
FROZEN	Sample ring stops writing; acquisition/diagnostics/protection continue independently.
EXPORT	Stream chronological frames plus metadata to reserved DDR slot; ring remains frozen.
COMMIT	Validate transfer counts/checksum and publish descriptor ownership.
RESUME	Restart ring and warm history; capture availability flags remain explicit.

This baseline is one sample bank. It cannot guarantee full records for arbitrarily close events while frozen/exporting. Campaign shot spacing shall account for readiness. Ping-pong buffering is optional and requires revised SRAM resource accounting.

ID	Requirement	Owner / verification
REC-001	Capture trips and selected warning/near-miss events, including benign decisions, with one primary event ID.	ES / S,H
REC-002	Do not freeze at the initial trigger and thereby lose post-event data. Preserve trigger-relative prehistory until post capture completes.	ES / S,H
REC-003	A secondary trigger cannot extend protection deadlines or silently replace the primary trigger. Log it in bounded metadata or count it as dropped.	ES / S,H
REC-004	Ring wrap, short prehistory, missing ADC frames, truncated posthistory and metadata overflow shall be explicit flags.	ES / S,H
REC-005	Recorder busy shall not stall protection. Failed capture shall remain observable via event-loss counters and minimal sticky cause/timestamp registers.	ES / S,H
REC-006	Loss of recording alone shall not silently grant benign continuation. Features used for continued operation must remain independently valid.	BOTH / S,H
REC-007	A nominal campaign shot lacking required record/scope evidence shall fail completeness or be separately reported; it shall not disappear from the accounting.	BOTH / C

12.3 DDR transfer and queue

Use PF_TPSRAM for local capture, a coherent streaming wrapper and verified CDC FIFO, CoreAXI4DMAController and appropriate interconnect, with a fabric-master FIC2 route to non-cached MSS DDR. DMA control uses a compatible AXI control route, proposed FIC0. Protection configuration/status uses FIC3/APB through CoreAPB3. Do not attach an AXI-Lite control port directly to APB without conversion. Sources: R04, R05.

ID	Requirement	Owner / verification
REC-008	Proposed DDR carveout is 64 MiB; physical addresses, all aliases and access rights shall be fixed consistently in HSS/firmware/device tree.	ES / I,H
REC-009	DMA shall access only the approved recorder region. Descriptor/payload lengths and address arithmetic shall be checked against boundaries.	ES / S,H
REC-010	Use descriptor ownership FREE -> WRITING -> COMPLETE -> READING -> FREE, with explicit error state. Publish COMPLETE only after data and required checksum are finalized.	ES / S,H
REC-011	Handle AXI backpressure/errors with bounded retry/timeout and no dependency in the protection path.	ES / S,H
REC-012	Address-cache aliases, memory ordering and uncached CPU mappings/cache maintenance shall be reviewed; FIC2 does not provide CPU cache coherence.	ES / I,H
REC-013	Reserve metadata/descriptors and alignment inside the queue; do not claim raw-payload capacity as guaranteed event count.	ES / A,H
REC-014	Full-queue policy shall increment loss counters and preserve existing complete events. Default is drop new waveform capture, retain minimal cause/time; alternate overwrite policy requires explicit mode/version.	ES / S,H

Calculated raw capacity is $\text{floor}(64 \text{ MiB} / 128,000) = 524$ payloads before metadata/alignment. Proposed allocation slot size is 160 KiB; at most 409 slots fit before queue-header reservations, so choose an actual slot count after descriptor sizing. A 400-slot candidate leaves approximately 1.5 MiB for queue headers and margin. This slot policy is a design choice, not a measured DDR result.

12.4 Persistence

Linux shall drain complete records to eMMC and the host. eMMC/SD multiplexing prevents an assumed independent simultaneous-storage path. DDR content is volatile; abrupt power failure can lose uncommitted and unarchived events. Store a committed-record index/checksum and report interrupted files after restart. Hold-up power/power-loss-proof capture is outside baseline scope.

13. MSS, AMP and ownership

Resource	Proposed owner
E51	HSS boot/monitor.
U54_1	Bare-metal safety configuration, arming commands, event IRQ service, DMA control and calibration management.
U54_2-4	Linux logging/archive/dashboard and campaign communications.
DAC and safety control bank	Bare-metal only.
Read-only status bank	Bare-metal and Linux through separate mapping/driver.
Recorder slot ownership	Fabric/DMA producer, bare-metal broker, Linux consumer under descriptor protocol.
Ethernet/eMMC/console	Reference-compatible owner allocation; freeze before integration.

ID	Requirement	Owner / verification
MSS-001	Validate the chosen HSS payload, hart mask and Linux/bare-metal combination on the installed official reference release.	ES / I,H
MSS-002	Define firmware, shared-message and Linux DDR regions with no overlap, including all CPU/fabric aliases.	ES / I,H
MSS-003	Enforce Linux restrictions on safety MMIO with the actual PMP/MPU/boot configuration and driver mappings. File permissions alone are not the hardware ownership mechanism.	ES / I,H
MSS-004	APB does not inherently identify the issuing hart. Document how access control is enforced before claiming isolation.	ES / I,H
MSS-005	Linux-to-bare-metal changes use versioned IHC/mailbox requests; validate range, state, sequence and transaction result.	ES / S,H
MSS-006	A request accepted for processing is not confirmation that the hardware configuration is applied. Return applied revision/status after completion.	ES / S,H
MSS-007	Event servicing shall be bounded; interrupts and queue activity shall not starve configuration health service. Hardware protection remains independent.	ES / S,H
MSS-008	Proposed bare-metal heartbeat every 10 ms, timeout 100 ms; freeze values against service behavior. Timeout revokes permission through hardware. This is distinct from the external fabric-clock watchdog.	ES / A,S,H
MSS-009	Failure to boot Linux shall preserve inhibit/protection and bare-metal diagnostics. Recover the connected viewer for the final target implementation.	ES / H

AMP is a supported platform capability (R11), but exact boot/peripheral/isolation configuration is OPEN. Root-level Linux restrictions and bus access shall be tested; no broad cybersecurity certification is implied.

14. Control/status register contract

14.1 Bus rules

The following offsets are proposed within a project-local APB control block. The physical base address is OPEN and shall be chosen to avoid reference-design conflicts. Registers are 32-bit aligned, little endian. RW = read/write while disarmed; RO = read only; WO = write command; W1C = write-one-to-clear qualified sticky non-safety status.

Writes shall have defined byte-enable policy, reserved bits shall read zero, and unsupported addresses shall return a defined error/default response without a hanging bus. If the chosen APB core lacks an error response, increment a reject counter and return documented benign read data. Do not clear safety state using a generic status W1C write.

Offset	Name / access	Proposed definition
0x000	DEVICE_ID / RO	Constant project identification, independent of firmware string.
0x004	SPEC_VERSION / RO	Register/event ABI revision.
0x008	BUILD_ID / RO	RTL build identifier or digest fragment.
0x00C	CAPABILITIES / RO	Implemented lanes, optional soft response/replay/test controls and recorder features.
0x010	COMMAND / WO	Validated command code; never a raw gate-enable bit.
0x014	STATE / RO	Protection state, armed/capture-ready/health bits.
0x018	TRIP_CAUSE / RO	Sticky bitset; first cause retained and concurrent causes accumulated.
0x01C	REJECT_COUNT / RO	Invalid/forbidden transaction count.
0x020	CONFIG_REVISION / RW	Shadow configuration revision.
0x024	CONFIG_COMMIT / WO	Atomic validation-and-apply request, disarmed only.
0x028	APPLIED_REVISION / RO	Revision actually active in fabric.
0x02C	CALIBRATION_ID / RW	Calibration identifier associated with the shadow set.

Offset	Name / access	Proposed definition
0x030	WARN_DEGLITCH_TICKS / RW	Proposed default 2, permitted range 0-2 ticks for current capture budget; larger values require a revised timing contract.
0x034	EMG_DEGLITCH_TICKS / RW	Default 0; allowed only 0-2 with approved evidence.
0x038	DECISION_MAX_TICKS / RW	Post-capture cap <=180 ticks for the proposed budget; further shortened by physical guard. Invalid larger settings rejected.
0x03C	RULE_ENABLE_MASK / RW	Only implemented/validated rules can authorize continuation.
0x040	MAX_FEATURE_AGE_TICKS / RW	Global cap; per-rule requirements may be stricter.
0x044	GUARDED_MAX_TICKS / RW	Maximum continued-operation validation interval; OPEN until envelope review.
0x048	CURRENT_GUARD_mA / RW	Software limit below immutable emergency ceiling.
0x04C	SLOPE_GUARD / RW	Proposed unsigned Q16.16 milliamps per 5 ns tick; ABI version identifies units.
0x050	PWM_PERIOD_TICKS / RW	Validated period.
0x054	PWM_DUTY_TICKS / RW	Validated duty below allowed maximum.
0x058	PWM_DEADTIME_TICKS / RW	Minimum approved deadtime; zero rejected in complementary operation.
0x05C	PWM_MAX_PULSE_TICKS / RW	Independent pulse cap.
0x060	MSS_HEARTBEAT_SEQ / WO	Monotonic/validated heartbeat service value.
0x064	MSS_TIMEOUT_TICKS / RW	Approved hardware health timeout; cannot disable protection while armed.

Offset	Name / access	Proposed definition
0x068	EXT_CLEAR_STATUS / RO	Requested/complete/failed state; actual external health verification.
0x06C	HEALTH_STATUS / RO	Observed health, clock lock and self-test summary.
0x070	SNAPSHOT_REQ / WO	Starts atomic status/timestamp snapshot.
0x074	SNAPSHOT_ACK / RO	Snapshot sequence/revision.
0x078	SNAP_TICK_LO / RO	Latched time low word.
0x07C	SNAP_TICK_HI / RO	Same snapshot high word.
0x080	EVENT_ID_LO / RO	Current/last event ID low.
0x084	EVENT_ID_HI / RO	Same snapshot high.
0x088	DECISION_STATUS / RO	Class, action, validity summary, rule ID.
0x08C	DECISION_LATENCY_TICKS / RO	Event-relative hardware decision time.
0x090	ADC_FRAME_COUNT_LO / RO	Snapshot of acquired frame sequence.
0x094	ADC_FRAME_COUNT_HI / RO	Same snapshot high.
0x098	ADC_ERROR_COUNT / RO	Transport/freshness errors.
0x09C	CAPTURE_LOSS_COUNT / RO	Events without complete waveform record.
0x0A0	RECORDER_STATE / RO	Warmup/ready/post/frozen/export/commit/error.
0x0A4	RECORDER_FLAGS / RO	Short pre, truncation, overflow, pending export.
0x0A8	DDR_FREE_SLOTS / RO	Validated queue free count.
0x0AC	IRQ_STATUS / RO	Pending source summary.
0x0B0	IRQ_ACK / WO	Acknowledge serviced events; does not clear protection latch.
0x0B4	ERROR_STATUS / RO	Config, rule conflict, arithmetic, clock and interface errors.

Offset	Name / access	Proposed definition
0x0B8	NONSAFETY_W1C / W1C	Clear specifically documented diagnostic sticky bits.
0x0BC	RESERVED / RO	Zero, retained for ABI expansion.

Calibration coefficients/rule tables require a separate indexed shadow window or memory-mapped block with length/version/checksum and an atomic apply handshake. Exact offsets and table layout are OPEN. DMA's own registers use its compatible AXI control interface, not these APB offsets.

14.2 Command, commit and interrupt policy

Proposed command codes are ARM 0xA5C30001, DISARM 0xA5C30002, CLEAR_TRIP 0xA5C30003 and SELF_TEST 0xA5C30004. These patterns prevent accidental generic writes; they are not authentication. ARM is accepted only from READY with valid configuration/calibration, healthy sources and capture/measurement prerequisites for the selected mode. DISARM revokes permission. CLEAR never arms.

CONFIG_COMMIT validates all shadow values, rule dependencies, checksums and margins while disarmed, then transfers one coherent snapshot to the protection domain. On success increment APPLIED_REVISION; on rejection leave the active configuration unchanged and record the reason. DAC analog updates/settling are additional to fabric commit and shall be included in the overall apply result.

Interrupt sources include primary decision, trip, capture complete, queue error and health change. Capture completion and the urgent protection notification may occur at different times. Event IDs correlate them. Linux reads mirrored RO status; bare-metal owns clearing/acknowledgement. A lost interrupt shall not hide the sticky cause or monotonic event counter.

15. Event record and file schema

Field	Proposed type / meaning
magic, schema_version, header_bytes	Recognizable format and versioned length.
event_id, campaign_id, shot_id	64-bit identifiers; campaign labels are ground truth, excluded from classifier.
rtl_build, firmware_build, software_build	Digests/version strings stored in manifest and record linkage.
rule_revision, calibration_id	The applied versions at event start.
trigger_tick, decision_tick, inhibit_tick	64-bit FPGA ticks; specify source and uncertainty.
first_cause, accumulated_causes	Preserve first observed source plus concurrent sources.
decision_class, requested_action, applied_action	Separate inference from actual hardware output.
feature_valid_mask, feature_age	Evidence validity and age per feature.
threshold_edge_records	Channel, level, polarity, observed tick, uncertainty/calibration reference.
adc_first_sequence, first_aperture_tick, sample_period_ticks	Raw waveform timebase; sample period candidate 200 ticks.
pre_count, post_count, trigger_frame_index	Actual counts, not requested lengths.
raw_channel_order, scale/offset data	I_dc, I_load, V_bus, V_sw; raw codes retained.
capture_flags, transport_errors, loss_count	Gaps, short history, busy/overflow/truncation.
local_backup_observed, watchdog/health	Backup status where observable; unknown is explicit.
payload_bytes, checksum	Bounded parser length and integrity check; algorithm version recorded.
scope_record_link	Independent physical measurement associated with event/shot.
envelope_revision, measured_initial_conditions	Voltage, current, switching phase and available temperature information.

Use a versioned binary waveform file plus human-readable JSON manifest or equivalent documented format. Parser limits shall reject excessive sizes/counts before allocation; incomplete files shall not appear as complete waveforms. A checksum detects accidental corruption, not malicious modification. Exact binary header packing/checksum polynomial is OPEN and shall be frozen with implementation tests.

16. Software and dashboard behavior

Component	Required user-visible behavior
Bare-metal service	Configure while disarmed, apply checked revisions, self-test, arm/disarm, service event IRQs and queue transfers.
Linux broker	Consume completed records without touching safety ownership; write archive and report health/loss counters.
Web dashboard	Event list, waveform plot, trigger/decision/inhibit markers, class/action/reason, validity, build/calibration IDs and completeness flags.
Campaign runner	Apply approved test plan, verify ready/health, execute interlocked shot, associate ground truth and scope record, enforce cooldown and preserve all outcomes.
Status API	Read health/queue/build state; numerical units and schema version included.
Command API	Request disarm/configure/self-test/arm through bare-metal; explicit accepted/applied/rejected states.

ID	Requirement	Owner / verification
SW-001	Dashboard shall distinguish design targets from measured values and illustrative traces from recorded data.	ES / I,H
SW-002	Display unresolved decisions and invalid/gapped records prominently; no interpolation presented as measured evidence.	ES / I,H
SW-003	A disconnected host/browser/network shall not alter protection thresholds, arming state or event deadlines.	ES / H
SW-004	Event polling/streaming shall be rate-limited so logging load cannot monopolize the control service.	ES / H
SW-005	Safety-changing requests require explicit operator action, appropriate authorization and state validation; no automatic arm from page load/reconnect.	ES / S,H
SW-006	Maintain a local UART console for health/configuration diagnosis independent of web availability.	ES / H
SW-007	Store actual apply acknowledgements, rejects and reasons in an audit log tied to revisions.	ES / H

The demonstration should make the physical response understandable first: event waveform, decision evidence, hardware action and baseline comparison. Implementation acronyms belong in developer views/documentation unless needed by the operator.

17. Calibrated replay and instrument requirements

Equipment	Capability to resolve before physical testing
Oscilloscope	Channels/probes sufficient for synchronized crossing, FPGA output, driver/gate and current/voltage measurements; bandwidth/sample rate/deskew suitable for reported uncertainty. Exact model OPEN.
Current and differential probes	Valid common-mode, bandwidth, current/voltage range and isolation for the actual test topology; loading documented.
Pulse/ramp source	Calibrated low-voltage signals spanning comparator slew/overdrive conditions and asynchronous phase.
Multichannel waveform source	Aligned channels and measured settling/bandwidth for replay. Tang Nano + DAC or lab AWG; exact update rate/resolution OPEN.
Bench source and energy fixture	Charging/current limit plus audited local energy, independent cutoff and safe discharge indication.
Gate hardware	Exact driver supplies/logic, inhibit behavior, retained device protection and optional profile selection.

Equipment access through grants/university labs is anticipated, not secured. PE owns booking/access confirmation, probe adequacy and hardware metrology; ES owns test automation and record association.

ID	TST-REPLAY requirement	Owner / verification
TST-001	Replay shall declare model revision, amplitude scale, time scale, offset, polarity, sensor bandwidth, clipping and injection location.	BOTH / I,H
TST-002	Scope the actual replay waveform at the injection point before scoring classification.	PE / H
TST-003	Injection after the amplifier tests only the downstream path; amplifier/sensor common-mode stress remains untested.	BOTH / I
TST-004	MV replay shall not be represented as MV insulation, device-temperature or short-circuit survival validation.	BOTH / I
TST-005	Resolution/update rate are measured properties of the selected source. A Tang Nano name alone or slow I2C DAC is insufficient.	PE / H
TST-006	Replay shots shall have a separate denominator from the required physical 1,000-shot campaigns.	ES / I,C

18. Verification plan and acceptance gates

18.1 Simulation and integration checks

Test ID	Scope	Pass condition
V-S01	Emergency priority under simultaneous soft/benign/config activity	Permission always revoked; emergency profile selected.
V-S02	FSM reset, illegal state and clear/rearm sequencing	No uncommanded gate enable; clear never arms.
V-S03	Deadtime/duty/pulse limits	Complementary outputs never overlap; invalid values rejected.
V-S04	Missing/same-tick/reversed/chattering threshold crossings	Correct validity flags; no unsafe arithmetic/benign authorization.
V-S05	Deadline under retriggers/interrupts/AXI stalls	Deadline never shifts later and no memory transaction stalls decision.
V-S06	Shadow commit and crossing coherency	Active configuration changes atomically while disarmed only.
V-S07	Fixed-point overflow, rounding and feature freshness	Conservative saturation/flags; golden model agreement.
V-S08	Ring wrap, short history, multiple triggers and metadata overflow	Chronology/counts/f lags correct.
V-S09	DMA ownership, range, full queue and timeout	No out-of-region write or premature COMPLETE descriptor.
V-S10	ADC previous-sample mapping and transport recovery	Correct aperture time, pipeline validity and gap accounting.
V-S11	Atomic 64-bit snapshots and counter wrap	No torn status; interval arithmetic remains valid.

Test ID	Scope	Pass condition
V-S12	Mailbox malformed/range/state requests	Reject/acknowledge safely; no direct arm/threshold bypass.

Use cocotb or equivalent custom-RTL tests, a numerical golden model and official vendor models for IP integration. Add random timing/phase and boundary cases tied to the requirements. Simulation does not prove analog delay, sensor behavior or power-stage survival.

18.2 Hardware gates

Gate	Required evidence before proceeding
G0 Access/tool	Actual lab access, selected instruments/driver, installed exact-part build/license and reproducible reference project.
G1 Wiring/logic	Revision-specific pin map, measured I/O rails, connector loopback, startup/default inhibit, external latch/watchdog tests with power switching disabled.
G2 Analog	Calibrated gain/offset/noise, threshold ordering, bandwidth/recovery and slew/phase timing intervals.
G3 Limited power	Reviewed fault path/energy bounds, one-shot interruption, gate/current/voltage limits and independent cutoff.
G4 Context	Development-set rules, overlap cases, conservative uncertainty response and closed decision timing.
G5 Capture/AMP	Raw-channel chronology, pre/post capture, DDR ownership, archive completeness, bare-metal/Linux split and stress tests.
G6 Campaign	Frozen envelope/calibration/rules/builds, preregistered held-out matrix and counter/evidence accounting.
G7 Final release	Clean rebuild, results/limits, disclosures, max-ten-page report and 3-5-minute working-hardware video.

18.3 Failure injection

Test ID	Injected failure	Required observation
V-H01	Bare-metal stall / lost heartbeat	Defined hardware timeout/inhibit; fixed emergency path remains active.
V-H02	Linux CPU, DDR, storage and Ethernet saturation	No protection-deadline extension; capture/queue losses counted.
V-H03	Stop or disturb fabric clock	External watchdog/local protection act; default/sticky inhibit persists.
V-H04	FPGA/device reset or rail sequencing	Both gates remain inhibited through startup/rearm.
V-H05	Comparator stuck high	Trip/check failure; no automatic rearm.
V-H06	Comparator stuck low / sense disconnect	Self-test/plausibility limitation documented; independent protection behavior measured.
V-H07	DAC/reference failure or wrong code	No raised fixed emergency limit; invalid evidence inhibits arming/continuation.
V-H08	ADC disconnect, stalled frame or clipping	Validity/freshness flags; affected rules cannot authorize continued operation.
V-H09	DMA stall, full queue, ring wrap/retrigger	Protection unchanged; explicit incomplete/loss record.

Test ID	Injected failure	Required observation
V-H10	Host/network loss	Autonomous protection and local event retention within declared capacity.
V-H11	Abrupt power loss	Gates default off; volatile/persistent-record limitations reported after recovery.
V-H12	Fault during soft response or clear request	Emergency set/profile dominates; no inadvertent switching permission.

19. Physical campaign and statistical contract

19.1 Freeze before testing

Freeze the operating envelope, topology, parts, calibration, rule constants, builds, test matrix, baselines, deadline definitions and exclusion policy before the held-out campaign. Hold out complete waveform/parameter families where feasible. Adjacent samples of a development waveform are not independent evidence of generalization.

Define every deadline-qualified fault as a valid commanded physical fault inside that preregistered envelope, with an allowable latest intervention measured from independently observed onset. A fault that does not cross a warning threshold remains in the denominator if its physical parameters qualify. Backup intervention cannot convert a missed Sentinel action into a success.

19.2 Matrix

Axis	Minimum specified coverage
Event family	Benign load step, controlled output fault, near-threshold ambiguous transient; diagnostics separately.
Initial condition	Multiple pre-event currents and bus-voltage bins within approved limits.
R/L/load parameters	Held-out combinations including overlapping initial slope and peak current.
Switching phase	Randomized/mapped phase including turn-on, turn-off, deadtime and freewheel states as applicable.
Sensor disturbance	Reviewed noise/slew/common-mode conditions and saturation recovery fixtures.
Temperature	Available approved bins; untested temperatures explicitly excluded from claims.
Software load	Nominal and high CPU/network/DDR/archive workloads.
Rule configuration	Tuned threshold baseline, threshold plus conventional blanking/hysteresis, contextual Sentinel; reviewed ablations.

Run at least 1,000 valid held-out physical load steps and 1,000 valid controlled physical fault pulses. Record aborted/invalid shots separately with reasons. A missing record or bad classification does not make a shot invalid retrospectively. Correlated cooldown/aging/drift shall be reported and monitored.

19.3 Outcome definitions

Outcome	Accounting
Benign accepted	Legitimate event continues inside limits with valid decision evidence.
Nuisance intervention	Any shutdown on a valid legitimate load step, including unresolved shutdown.
Correct fault action	Sentinel's own action meets the fault's preregistered intervention deadline and physical limits.
Missed/late fault	Sentinel fails to act within the deadline, even when backup protection saves the hardware.
Unresolved fault shutdown	May be a successful safe action; counts as an abstention for diagnostic label accuracy.
Incomplete evidence	Capture/scope/manifest missing; count and report separately alongside physical outcome.
Invalid shot	Independently defined instrument/fixture condition outside preregistered validity; reason retained.

Report confusion matrix, diagnostic abstention, physical action success, record completeness and timing/stress distributions independently.

For zero errors in 1,000 independent Bernoulli trials, the one-sided 95% upper rate bound is $1 - 0.05^{(1/1000)} =$ approximately 0.299%. It pertains to the sampled distribution and independence assumptions. At least 2,995 zero-error independent trials are required for a corresponding bound below 0.1%. Aggregate bounds do not automatically apply to each parameter bin.

20. Baseline comparison and ablation

Baseline	Required fairness
B0 Practical fixed threshold	Sweep threshold using the same sensors, emergency ceiling, plant, driver action and campaign envelope.
B1 Threshold with conventional qualification	Tune reasonable hysteresis/blanking/deglitch within identical physical limits; disclose parameters and delays.
B2 Contextual Sentinel	Add calibrated timing/context rules; unchanged emergency authority and physical hardware.
A0 No slope	Remove threshold-interval feature while keeping other observations and limits.
A1 No extra context	Remove topology/PWM/current-relation features; show what slope alone contributes.

Compare nuisance intervention against missed/late fault rate, intervention latency, peak current, peak V_{DS} and integrated device stress. Match fault-detection performance or present a tradeoff curve when exact matching is unavailable. Select baseline tuning on development data; do not tune on held-out outcomes.

Commercial programmable gate drivers already provide tailored switching and diagnostics. The novelty claim is the tested combination of bounded uncertain decisions, parallel observations and an inspectable evidence record. Sources R14/R15 support the need for a modest claim. No patent-novelty conclusion or universal MCU superiority is asserted.

21. Resource, power and tool constraints

Resource	Planning ceiling / available denominator	Status
Logic elements	25,000 / 254,000 = 9.84%	Proposed whole implemented fabric allowance including integration; verify reference footprint first.
18 x 18 math blocks	32 / 784 = 4.08%	Mapping depends on chosen arithmetic width/parallelism.
20 Kbit LSRAM blocks	80 / 812 = 9.85%	Ring, validity, metadata, FIFOs and packing overhead must fit.
Header signal occupancy	Full 26/26; reduced 24/26	Connector feasibility, not package-wide I/O utilization.
Raw recorder input	8 MB/s	Additional metadata /descriptor/export overhead budgeted separately.
Event payload	128,000 raw bytes + metadata	Requested 16 ms event with continuous 1 MSPS cadence.
DDR carveout	64 MiB	Queue format/slot count/addressing OPEN.

The raw ring alone needs at least 52 nominal 20 Kbit blocks by bit capacity; real width/depth packing or ECC can require more. An 8 KiB validity map adds at least four nominal blocks. The total 80-block allowance leaves a finite metadata/FIFO/integration margin and shall be revised after actual mapping. Double-buffering 128 KiB rings will exceed this baseline memory allowance before other buffers are counted.

ID	Requirement	Owner / verification
SYS-009	Report total and incremental resources relative to the same reference build, plus routed timing and unmet constraints.	ES / I,S
SYS-010	Validate exact MPFS250T-FCVG484E tool/license support and required IP before board work; Silver kit-part support is not universal MPFS250 support.	ES / I,S

ID	Requirement	Owner / verification
SYS-011	Record tool/IP/reference versions and a clean reproducible build path.	ES / I,S
SYS-012	Measure equivalent-workload power where equipment permits. Kit rail readings/whole-kit input include DDR/Ethernet/software loads and shall be labeled accordingly.	BOTH / H

Sources R02, R05, R12 and R13 cover platform resources, DMA and licensing. No synthesis result is available at this revision.

22. Risk register and release decisions

Risk ID	Failure / consequence	Mitigation / release evidence	Owner
K01	Fault/benign signatures overlap	Real extra observations, uncertainty intervals, conservative expiry; held-out overlap evidence.	BOTH
K02	Comparator jitter/noise produces misleading slope	Qualified fast candidate, measured slew/phase uncertainty, hysteresis and bounded rejection.	PE
K03	Amplifier offset/saturation/common-mode failure	Calibrated residual error/recovery, valid input topology, retained local protection and explicit common causes.	PE
K04	Fault branch bypasses protected switch	Current-path drawing and measured source-current interruption.	PE
K05	Rapid turn-off exceeds voltage/energy limits	Profile/clamp/layout review and matched-current physical tests.	PE
K06	Wrong bank/routing/header loading	Delivered-revision schematic/PDC sign-off and loopback/timing measurements.	BOTH
K07	All 26 header positions occupied	Off-kit fault/load controls release two positions; no implicit spare signals.	BOTH
K08	FPGA/processor/clock failure	External sticky trip, clock watchdog, default inhibit and hardware deadlines.	BOTH
K09	Linux accesses safety MMIO	Actual PMP/MPU/boot/peripheral ownership checks and bypass tests.	ES
K10	DMA/cache/queue corruption	Reserved regions, ownership protocol, checksums and error injection.	ES
K11	Recorder retrigger/power loss	Explicit busy/truncation/loss flags; volatile/persistence limits stated.	ES
K12	Lab access/parts delayed	Confirm equipment bookings/BOM early; logic/replay milestones before kit arrival.	PE
K13	Linux/AMP integration slips	Hardware and bare-metal first; small connected viewer; avoid unrelated features.	ES
K14	Proposal deadline/naming wrong	Signed-in portal verification; plan for earlier supplied deadline until confirmed.	BOTH

23. Development plan and responsibilities

Dates	Deliverable / gate	PE responsibility	ES responsibility
Oct 2-3, 2026	Proposal, rules/portal checks, diagram and roles	Review physics/plant commitments	Review kit/interfaces/tool commitments
Oct-Nov pre-work	G0-G2 preparation	Equipment access, device/driver selection, sensing/fault design	Exact-part reference build, RTL golden model, ownership plan
Nov 30-Dec 18	Kit logic and measured emergency path	Gate/default-inhibit fixture and timing	Loopback, capture logic, bare-metal/AMP skeleton
Dec 19-Jan 17, 2027	Calibrated sensing and limited physical power	AFE/energy/current path and physical measurements	Bounded rules and recorder integration
Jan 18-Feb 7	Capture/export and connected viewer	Context validation and stress measurements	ADC transport, DDR queue, Linux archive/dashboard
Feb 8-Mar 7	Held-out campaigns, baselines and failures	Physical matrix, metrology, optional soft-profile validation	Automation, outcome accounting, ablations, software stress
Mar 8-20	Feature freeze and reproducibility	Hardware release evidence and physical demo	Clean build, event viewer and report evidence
Mar 21-27	Final checks/submission	Final measurement/limitations review	Files, disclosures, links, build instructions/video

No feature shall be withheld merely to manufacture final-round evolution. Refinement should follow measurements. Both members review classifier physics and implementation; shared FPGA ownership requires one versioned integration baseline and explicit module ownership to avoid incompatible concurrent changes.

24. Block diagram specification

The submission architecture shall be technically explicit and readable. Page one communicates the plant, signal path, protection authority, FPGA contribution, recorder and connected evidence. Page two carries implementation details that establish feasibility.

Main architecture page	Required granularity
Title	Project, registered team, Track 2, kit/part, revision/date; draft status if unresolved.
Deployment inset	MV/SST/800 VDC/racks with 'target deployment, not built' label.
Plant	48 V source, audited energy concept, SiC half-bridge, interruptible output fault R/L and load-step branch.
Sensing	Two different current paths; three thresholds per path; four-channel ADC; DAC/fixed references.
Parallel paths	Red emergency latch/local backup; contextual timestamp/rules path; emergency priority explicit.

Main architecture page	Required granularity
Response	Continue, validated shutdown or unresolved shutdown; optional soft profile identified.
Fabric boundary	Clock target, major IP/custom RTL roles, hardware deadlines, recorder SRAM/DMA.
MSS boundary	E51 HSS, bare-metal U54_1, Linux U54_2-4; controlled configuration ownership and Ethernet.
Interfaces	Counts/logic/rates rather than every individual wire: six comparator inputs, ADC serial lanes, I2C, isolated control, FIC2/FIC3.
Results panel	Targets and timing origins; tuned baseline and 1,000/1,000 held-out campaign intent.
Credibility panels	Three-to-five risks, dated gates/role owners, explicitly estimated resources/data rate.
Legend	Off-kit hardware, vendor IP, custom RTL, software; isolation boundaries and emergency priority.

Page two shall contain candidate J26 allocation with routing status, exact IP/interface roles, memory/record sizing, latency conditions and capture limitations. A full schematic pinout is not required on page one. Precise pin numbers shall not imply released verification. Update the earlier diagram's 24/26 statement to distinguish the full 26-position and reduced 24-position allocations before publishing a new revision.

Original vector artwork is preferred. Never present illustrative waveforms as measured results. A small illustration of overlapping peak/slope events can explain the experimental question, but its labels must identify it as illustrative. Technical terms are retained when meaningful; peripheral implementation details belong in the notes.

25. Contest compliance and communication

The supplied rules govern the competition; this specification records them as requirements relevant to the team. It does not instruct an assistant to register, publish, contact organizers or accept terms.

Requirement	Source / implementation
Team	Two or three eligible members; one team per participant; registered identity and eligibility checked by team.
Track/platform	Exactly one track; Track 2 and provided kit selected.
Proposal fields	Overview <=300 words; innovation <=250 words; platform selection; architecture PDF.
Proposed wording	Existing revised overview 271 whitespace words; innovation 219. Portal's actual counter remains controlling.
Supplemental sharing	One public shared folder/drive link via allowed service; accessible throughout competition; no password; portal filename convention.
Submitted file freeze	Do not modify/replace/remove submitted files after deadline without organizer authorization. Keep later work separate.
Language	English submission and supporting materials.

Requirement	Source / implementation
Third-party material	Permission/license/disclosure obligations reviewed; protected images are not universally forbidden if express written permission exists.
Material AI assistance	Disclose ChatGPT/Codex contribution; team engineering understanding and validation required.
Final package	Maximum-ten-page project documentation; source/project files; README/build instructions; 3-5-minute demonstration; IP/open-source/AI disclosures.
Ownership/confidentiality	Supplied Section 11 retains stated ownership while granting broad licensing and describing non-confidential submission; team reviews terms.

Supplied PDF deadline: October 3, 2026 at 11:59 PM PDT, equivalent to October 4 at 2:59 AM EDT. The current official contest page says October 24 for proposal submission; signed-in portal deadline/current rules revision/naming remain OPEN. The schedule may have been extended. Plan against the earlier date until confirmed. Supplied final deadline: March 27, 2027 at 11:59 PM PDT, equivalent to March 28 at 2:59 AM EDT. Source R01 and supplied rules Section 5.

Proposal scoring: feasibility 35%, innovation 25%, clarity 20%, impact 20%. Final scoring: implementation 35%, innovation 25%, video 20%, documentation 20%. Risks, interfaces, IP and development plan are feasibility evaluation factors; extra standalone form fields are not assumed. These justify including concise engineering notes in the architecture PDF.

Communication shall claim measured selectivity in the defined prototype envelope, reproducible evidence and potential converter-developer usefulness. Reduced AI training interruptions and broader applications are potential downstream impacts, not demonstrated contest results. Nonvolatile configuration does not replace startup interlocks; configuration SEU immunity does not imply all state/memory/sensing is immune. A first-place award cannot be guaranteed from a specification.

26. Open-decision register

Decision	Closure evidence	Owner / gate
O01 Team identity and eligibility	Registered team/member details confirmed	BOTH / proposal
O02 Portal deadline/naming	Signed-in current instructions/current rules	BOTH / proposal
O03 Equipment access	Actual booking/grant/lab availability and instrument models	PE / G0
O04 Exact kit revision and constraints	Schematic + package/net/rail table + loopback	BOTH / G1
O05 Gate driver and emergency profile	Part/delay/polarity/protection and measured gate/stress behavior	PE / G3
O06 AFE current topology/amplifier	Complete current loops/common-mode, offset/noise/recovery calibration	PE / G2
O07 Thresholds/reference/error margins	Hardware limits and uncertainty intervals	BOTH / G2-G4
O08 External latch/clear/watchdog	Circuit, set dominance, heartbeat/timeout and startup tests	BOTH / G1
O09 ADC drivers/protocol/reset	Differential input transfer, dual-lane setup, timing, skew and recovery plan	BOTH / G2,G5
O10 Rule table/feature age/guard	Development-set evidence plus physical time/energy bounds	BOTH / G4
O11 Oscillation/sensor diagnostics	Frequency/window and detectable failure modes	BOTH / optional
O12 200 MHz/resources	Routed timing, CDC review and actual block mapping	ES / G4-G5
O13 AMP/peripheral/access protection	Tested HSS/hart/memory/PMP/MPU configuration	ES / G5
O14 DDR addresses/slots/event ABI	Coherent aliases, descriptor layout, checksum/parser tests	ES / G5
O15 Physical test limits	Frozen envelope, pulse/cooldown/metrology and independent cutoff	PE / G3,G6
O16 Held-out matrix/baselines	Preregistered families, tuning freeze, outcome rules	BOTH / G6
O17 Soft turn-off	Validated profile setup and stress; otherwise feature absent	PE / optional
O18 Replay source	Measured resolution/cadence/channel alignment; otherwise lab fixture only	PE / optional

Every closure shall record date, reviewer, source/build/calibration revision and supporting file. A closed design choice can reopen when hardware changes; affected tests must then be repeated.

27. Acceptance matrix and release package

Acceptance area	Required evidence
Protection authority	V-S01/S02/S03/S05; G1/G3; V-H03/H04/H12.
Analog/slope integrity	G2; V-S04/S07; independent slew/phase/common-mode calibration.
Contextual selectivity	G4/G6; overlap cases, bounded ambiguity, held-out baselines/ablations.
Timing/stress	Qualified threshold/inhibit and physical action measurements; uncertainty, peaks and energy.
Recording	V-S08/S09/S10/S11; G5; missing/retrigger/overflow/power-loss outcomes.
AMP/software ownership	V-S06/S12; G5; Linux bypass/stress/disconnection tests.
Reproducibility	Tool/IP/build IDs, exact constraints/BOM/firmware, clean build and event parsing.
Contest materials	Validated claims, diagram/naming/link, team identity, final report/video and disclosures.

Release package shall contain RTL/testbench/golden model; SmartDesign/project and constraint files; firmware/HSS/Linux configuration; AFE/power schematics and candidate-to-final BOM; calibration files; register/event ABI; campaign plan/results and scope associations; build/programming instructions; user arming/disarming/test procedure; limitations/common-cause record; and third-party/open-source/AI provenance.

The demo shall show a benign event, a comparable fault, an ambiguous event, and software failure behavior. Start with gate/current evidence and finish with quantitative results and reproducibility. Show the dashboard as the record explaining hardware behavior, not as the main accomplishment.

28. Primary-source register and provenance

Manufacturer facts are limited to the scope explicitly cited; surrounding equations, architecture choices and proposed requirements are engineering analysis. Links were checked during this conversation on October 2, 2026; mutable reference-project pages must be pinned to a commit for implementation. Sources do not prove a completed prototype.

ID	Source	Use
R01	Official contest page	Current schedule/track context; differs from supplied PDF deadline.

ID	Source	Use
R02	Icicle user guide	Kit interfaces/memory and revision-specific review.
R03	Official Icicle reference design	J26 routes, candidate physical positions and reference build.
R04	MSS technical reference	FIC/APB/DDR/ownership design.
R05	CoreAXI4DMAController	Stream-to-memory capability and license context.
R06	MCP6561 family datasheet	Conditional delay; discarded universal 50 ns assumption.
R07	TLV3501/TLV3502 datasheet	Faster comparator candidate and conditional timing.
R08	OPA356	Amplifier candidate; bandwidth/offset tradeoff.
R09	AD8410A	Conventional current-sense bandwidth example.
R10	ADS9224R datasheet	ADC/input/serial transport candidate.
R11	PolarFire SoC software development guide	AMP/HSS implementation context.
R12	Fabric resource table	Resource allowance denominators.
R13	Libero licensing	Exact Icicle kit-part Silver support.
R14	62EM1 programmable driver	Prior capability context for modest novelty claims.

ID	Source	Use
R15	Microchip SiC characterization	Device withstand is test-condition dependent.
R16	ADuM110N datasheet	Isolation candidate, delay/default behavior subject to selected variant/conditions.
R17	MCP4728 documentation	Quad DAC/reference and EEPROM context.
R18	Microchip 700 V SiC candidate family	Candidate device family; exact selection OPEN.
R19	NVIDIA 800 VDC architecture	Application motivation; no prototype deployment claim.

Supplied primary rules source: PolarFire_FPGA_Design_Contest_2026_Offical_Rules.pdf, workspace root; Sections 2-9 and 11-14. Previous local working artifacts: SST_Sentinel_Adversarial_Review.md; SST_Sentinel_Proposal_Text.md; SST_Sentinel_Architecture_Review_Draft.pdf.

Material AI assistance: OpenAI ChatGPT/Codex assisted with rule/source review, adversarial architecture refinement, requirement drafting, calculations, interfaces, register/event proposals and document production. Team understanding, implementation ownership, component verification, physical validation and truthful reporting remain required. The team shall amend the contest disclosure to reflect actual completed engineering work.